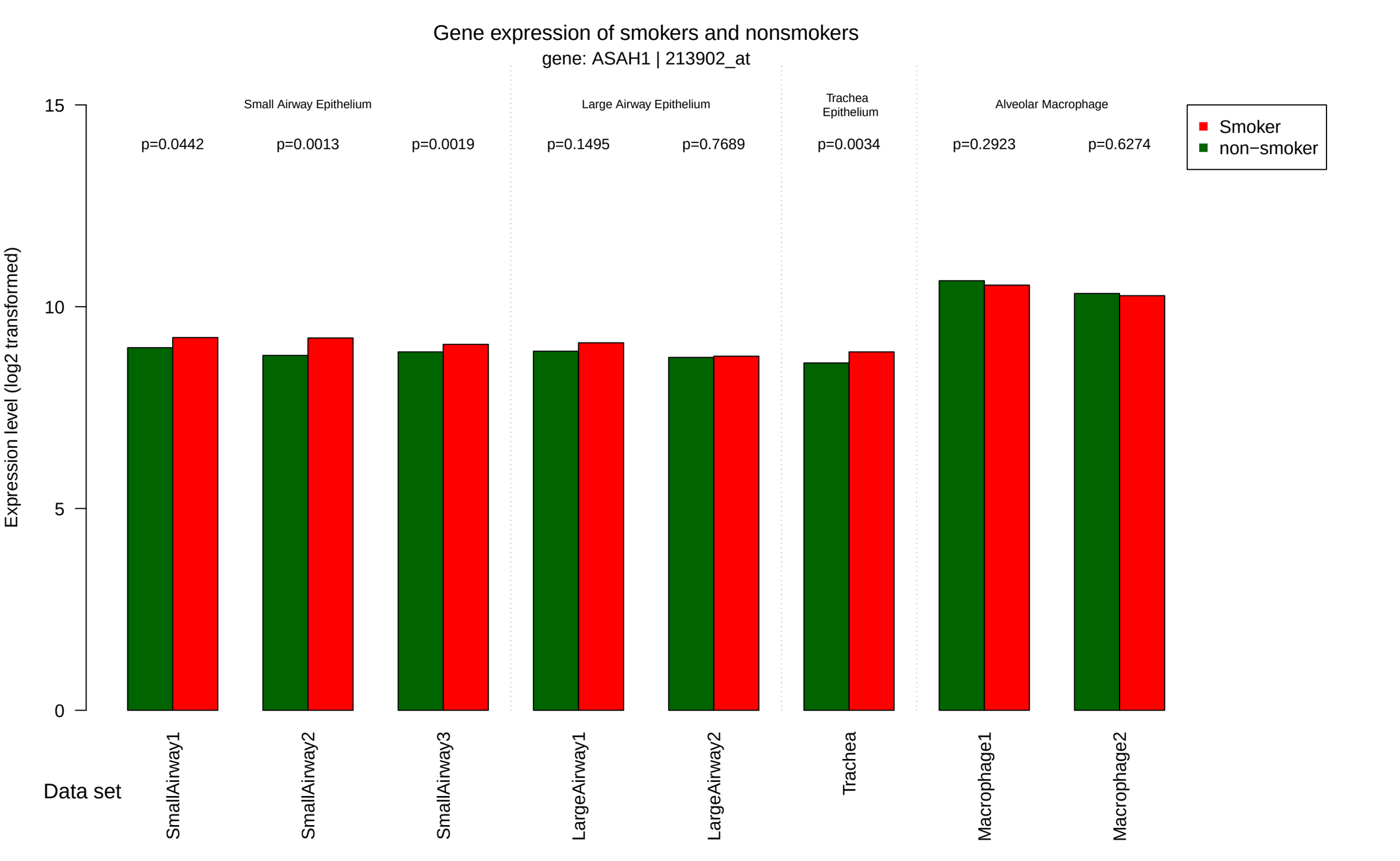
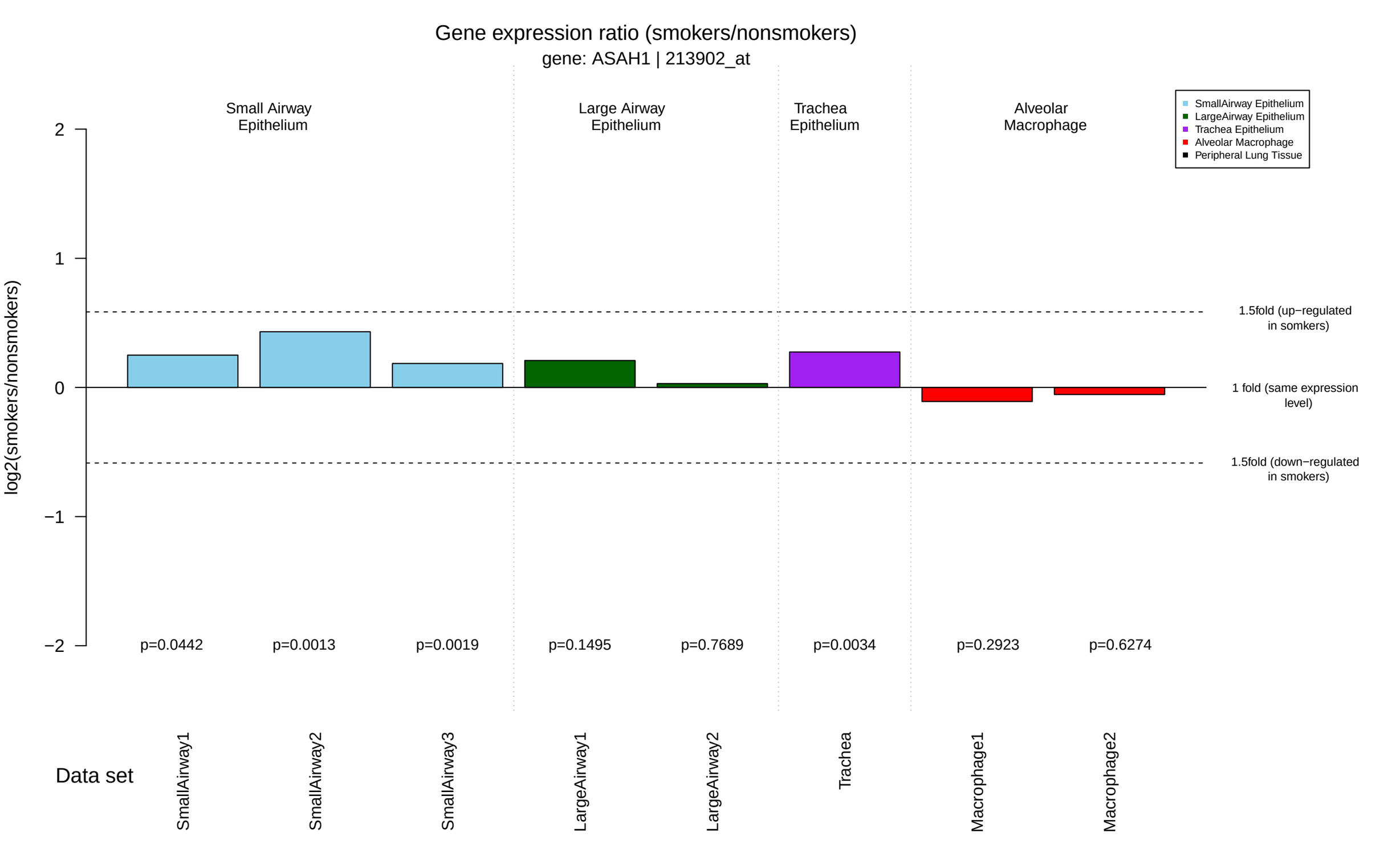
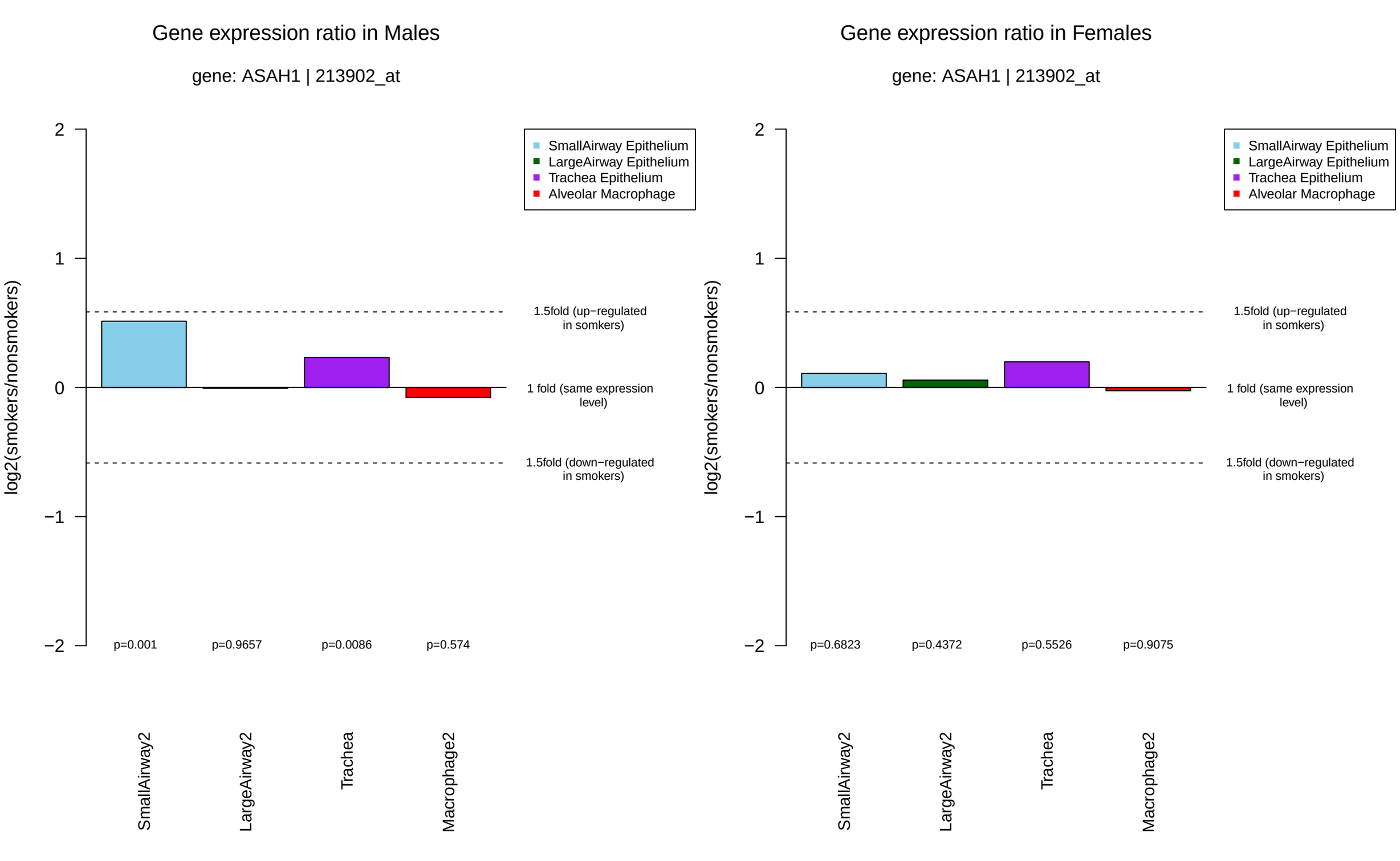
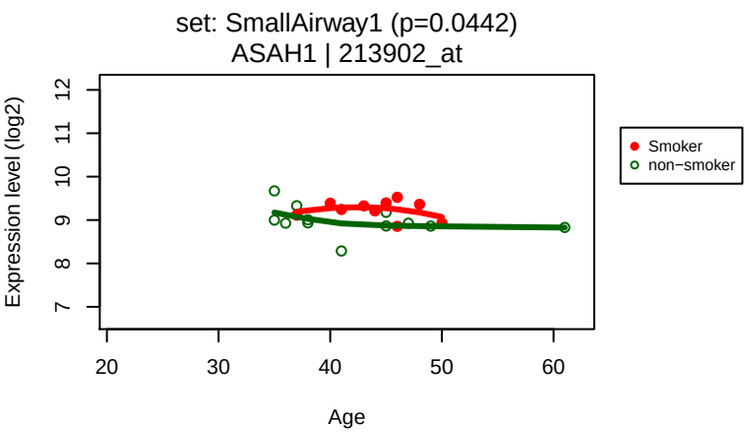
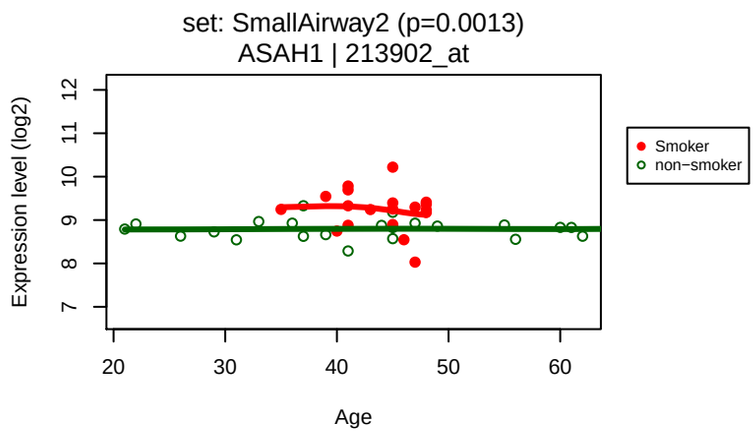
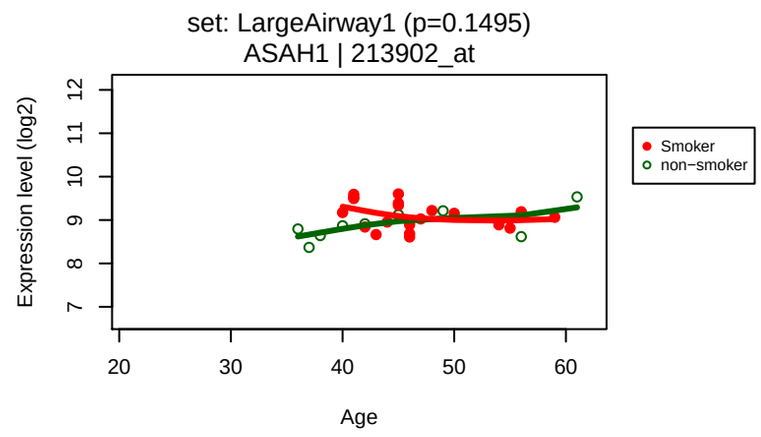

Gene expression of smokers and nonsmokers
gene: ASAH1 | 213902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0442
p=0.0013
p=0.0019
p=0.1495
p=0.7689
p=0.0034
p=0.2923
p=0.6274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

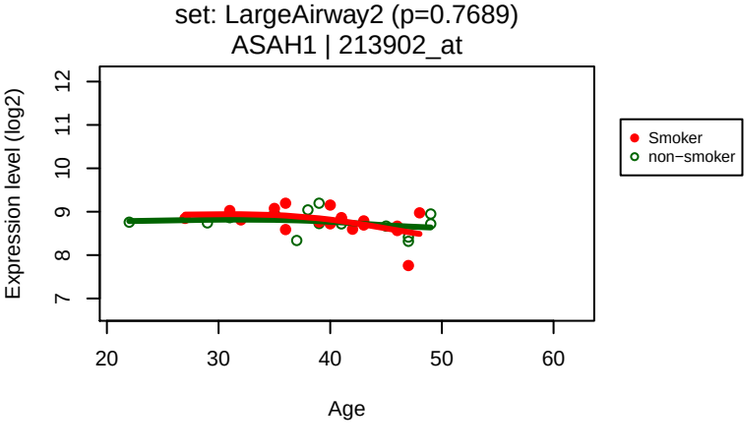

Gene expression ratio (smokers/nonsmokers)
gene: ASAH1 | 213902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0442
p=0.0013
p=0.0019
p=0.1495
p=0.7689
p=0.0034
p=0.2923
p=0.6274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

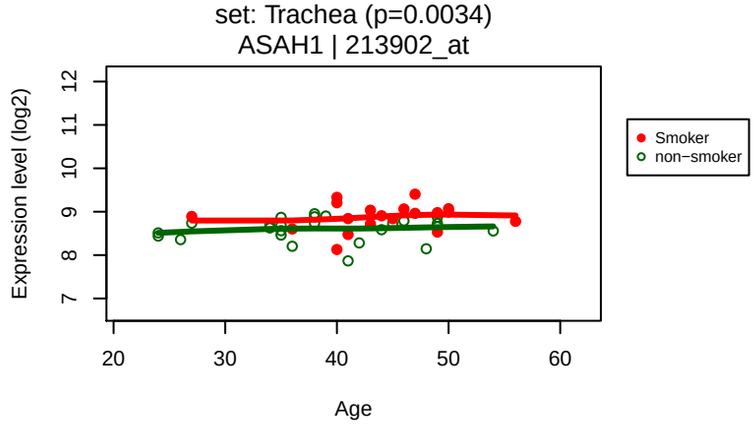

Gene expression ratio in Males
gene: ASAH1 | 213902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.001
p=0.9657
p=0.0086
p=0.574
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ASAH1 | 213902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6823
p=0.4372
p=0.5526
p=0.9075
SmallAirway2
LargeAirway2
Trachea
Macrophage2

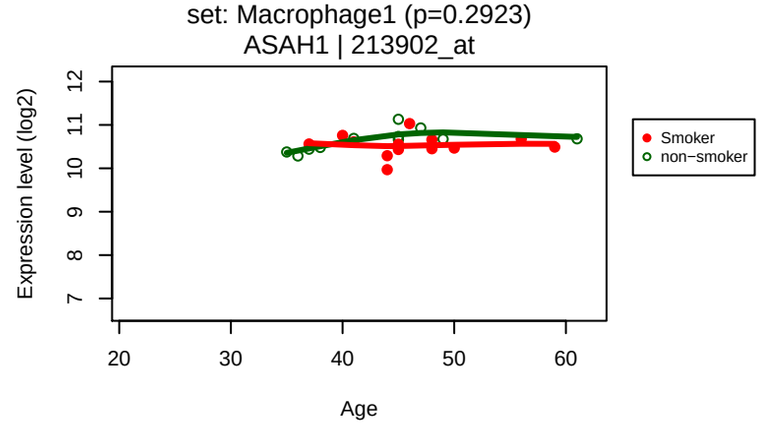

set: SmallAirway1 (p=0.0442)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker

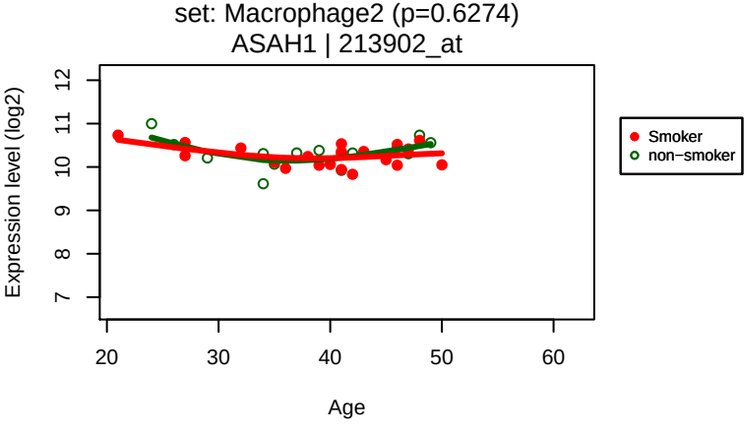

set: SmallAirway2 (p=0.0013)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker

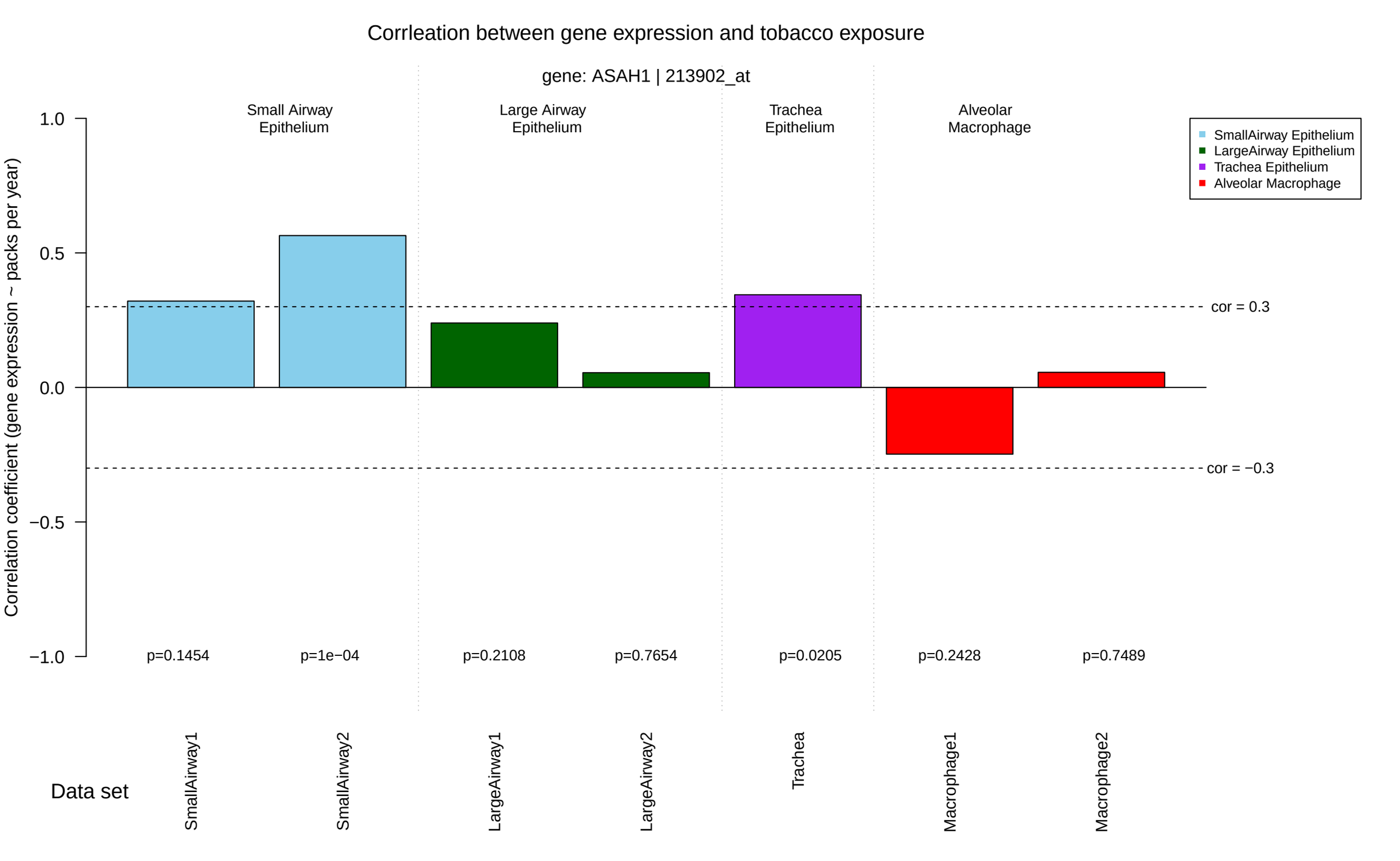

set: LargeAirway1 (p=0.1495)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7689)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0034)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2923)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.6274)
ASAH1 | 213902_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ASAH1 | 213902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1454
p=1e−04
p=0.2108
p=0.7654
p=0.0205
p=0.2428
p=0.7489
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2